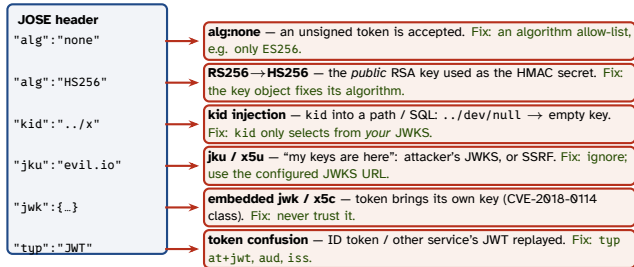


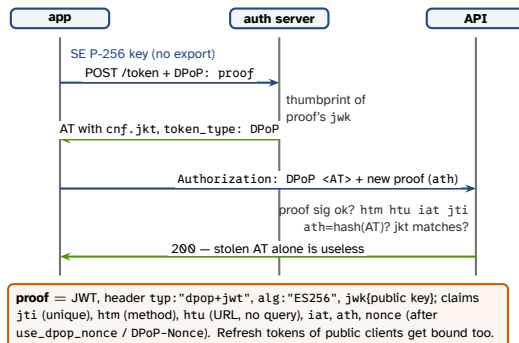
In one line: Every classic JWT break is the verifier **trusting the token to describe how to verify itself**: the header's alg, kid, jku, jwk are *attacker input*. Fix: the server decides algorithm, key and token type (RFC 8725). A signed bearer token is still stealable — **DPoP** (RFC 9449) binds it to a key the client holds, and the **OAuth Security BCP** (RFC 9700, Jan 2025) requires public clients' refresh tokens to be **sender-constrained or rotated**.

Every header field is attacker input



Payload: a weak HS256 secret fails to offline brute force of one captured token (hashcat -m 16500); RFC 7518 wants ≥ 256 random bits. Java SE 17/18 accepted ECDSA r=s=0 (CVE-2022-21449).

DPoP — a token only its key holder can use



Validation that cannot be steered (RFC 8725)

- Algorithm from config, not from the token** (§3.1): the key object carries its one algorithm; asymmetric keys can't be used as HMAC secrets.
- Keys from your trust anchor**: JWKS from the issuer's configured URL, cached; kid only *selects*. Ignore jku, x5u, jwk, x5c unless pinned to an allow-list.
- Explicit typing** (§3.11): typ: at+jwt (RFC 9068) for access tokens, dpop+jwt for proofs; mutually exclusive rules per token kind (§3.12) stop cross-JWT confusion.
- Claims**: iss exact, aud = me, exp/nbf with small leeway, sub format; crit header you don't understand ⇒ reject.
- Received claims are input** (§3.10): kid, sub, email into SQL, paths, LDAP, logs = injection.

JWS vs JWE

JWS (RFC 7515, 3 parts): integrity + origin, payload *readable*.
JWE (RFC 7516, 5 parts): header . encrypted key . IV . ciphertext . tag): confidentiality; alg wraps a CEK (RSA-OAEP, ECDH-ES, dir), enc seals (A256GCM). Avoid RSA1_5 (Bleichenbacher oracle); validate ECDH-ES points (invalid-curve attack). Need both? **sign, then encrypt** (nested JWT, cty: "JWT"). A JWE made with the server's *public* key proves nothing about who made it.

Example — a verifier the token cannot steer (jose, Node)

```
const JWKS = createRemoteJWKSet(
  new URL('https://auth.example.com/.well-known/jwks.json')
)
const { payload } = await jwtVerify(token, JWKS, {
  algorithms: ['ES256'], // allow-list: none / HS256 fail
  issuer: 'https://auth.example.com',
  audience: 'https://api.example.com',
  typ: 'at+jwt', // explicit typing, RFC 9068
  clockTolerance: 30, // seconds of skew
  maxTokenAge: '15m', // iat bound, even if exp is far
})
```

Test your own API — each must be refused

alg none/None · RS→HS with the public key · kid path/SQL · jku to your host · embedded jwk · signature stripped · expired / future nbf · ID token or other aud · refresh token reused after logout.

RFC 9700 — the rules

PKCE	public clients MUST ; AS MUST reject a code_verifier with no challenge (downgrade)
implicit / ROPC	implicit SHOULD NOT ; password grant MUST NOT
redirect URI	exact string match (except localhost port)
refresh tokens, public client	MUST be sender-constrained (DPoP / mTLS) or rotated
access tokens	SHOULD be sender-constrained; least privilege, audience-restricted

Rotation detects theft *after* both parties use the RT (reuse → revoke family); **sender-constraining** makes a stolen RT *unusable* without the private key.

Tokens on the phone

iOS: Keychain, ThisDeviceOnly; DPoP key in the **Secure Enclave** (SecureEnclave.P256.Signing) — signs, never exported, so a backup/jailbreak dump does not carry it. **Android**: Android Keystore. **RN**: react-native-keychain / expo-secure-store — **never** AsyncStorage (plain file). Access token in memory; log out = revoke RT + wipe + drop key.

Traps

- ! "We verify the signature" — with which algorithm and key? If the token chose either, it's forgeable.
- ! decode() vs verify(): reading claims without verifying is an auth bypass.
- ! HS256 shared across services: every service can mint tokens for all the others.
- ! JWE is not authentication — anyone with your public key can encrypt to you.
- ! DPoP without jti/iat checks: a proof can be replayed within its window; use nonces.
- ! Blocking alg:none by string compare: "None" slips through (*unverified*) — allow-list instead.

Remember

"The server picks alg, key and type; the token only picks which of my keys (kid). Bind what you can't keep secret."